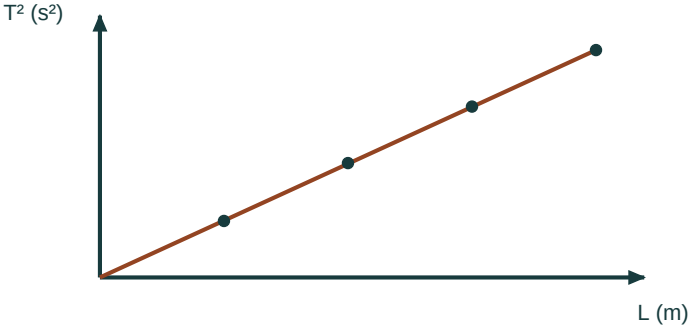
05

Mechanics and Heat Lab

A field guide to understanding, working and checking.

Pendulum: choose axes that give a straight line



Slope = change in T² / change in L

Slope = 4 pi² / g

So g = 4 pi² / slope

Use small-amplitude oscillations.

All distinct experiments in the folder: pendulum, free fall, static and dynamic springs, force table, cooling, Boyle law, expansion, specific heat, latent heat and Joule equivalent. Duplicate versions are consolidated.

12 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Earn marks from tables, graphs and uncertainty
2. Simple pendulum
3. Free fall and axis choice
4. Static Hooke law
5. Dynamic spring and gravity
6. Force table and equilibrium
7. Newton cooling law
8. Boyle law
9. Linear thermal expansion
10. Specific heat, latent heat and Joule equivalent
11. Calorimetry: determine a solid specific heat
12. Latent heat and electrical-to-thermal conversion

LESSON 01 / 12

Earn marks from tables, graphs and uncertainty

After this lesson, you can

- Calculate a fitted-line slope with the correct axis units.
- Distinguish percentage error from measurement uncertainty.

Start with: Coordinate differences; division; raw and derived measurements.

THEORY

A lab result is a measured quantity with units and a justified method. Record raw observations before calculating derived columns. A slope is change in the vertical variable divided by change in the horizontal variable. Its units are essential and determine how to recover the desired constant.

RULE

$$m = \Delta y / \Delta x$$

$$\% \text{ error} = 100|x_{\text{measured}} - x_{\text{reference}}|/|x_{\text{reference}}|$$

m slope of the selected vertical variable against horizontal variable (vertical unit / horizontal unit)

x_{measured} experimental estimate (same as reference)

$x_{\text{reference}}$ accepted comparison value (same as measurement)

A method that earns marks

1. Record raw readings with units before calculating derived quantities.
2. Label axes with quantities and units and draw a suitable best-fit line.
3. Choose two widely separated points on that fitted line and calculate rise over run.
4. Use axis units to interpret the slope; compare with a reference using the stated percentage-error formula.

WORKED EXAMPLE

A fitted line rises by 0.8 square seconds over 0.2 m. Its slope is 4 square seconds per metre. It is not 0.25 m per square second unless the axes are reversed.

1. The example has vertical change 0.8 s^2 and horizontal change 0.2 m.
2. Slope is rise over run: 0.8/0.2 = 4 s^2/m.
3. The reciprocal, 0.25 m/s^2, belongs to a graph with reversed axes and is a different slope.
4. A slope should come from the fitted line, not necessarily from two adjacent raw observations.

Common mistake
Do not force a line through the origin unless the model and measurement offsets justify it.

Exam habit: Write slope units next to the numerical calculation; they help reveal an inverted axis choice.

SELF-CHECK

Can percentage error alone describe repeatability of a measurement?

Answer: No. It compares with a reference; repeated-reading spread or an uncertainty estimate is needed to discuss repeatability.

Uncertainty is different from percentage error

An uncertainty describes measurement quality. Percentage error compares a result with a reference value.

- Use the uncertainty method specified by your experiment or instrument; do not invent a precision from extra calculator digits.
- Give the uncertainty the same unit as the value. Round the value to the same decimal place as the reported uncertainty.
- Repeated measurements help estimate and reduce the effect of random scatter. They do not remove a consistent zero offset.
- An uncertainty range is not a guarantee that every possible error is inside it.

reported result $=$ value $\pm$ uncertainty

$$\% \text{ error} = 100\frac{|x_{\text{measured}} - x_{\text{reference}}|}{|x_{\text{reference}}|}$$

A result is $(9.6 \pm 0.2)\,\mathrm{m/s^2}$, and the reference is $9.8\,\mathrm{m/s^2}$. State the given uncertainty and calculate percentage error.

1. The stated uncertainty is $0.2\,\mathrm{m/s^2}$.
2. Percentage error is $100|9.6 - 9.8|/9.8 \approx 2.04\%$.

Uncertainty: $\pm 0.2\,\mathrm{m/s^2}$. Percentage error: about 2.04%. They answer different questions.

LESSON 02 / 12

Simple pendulum

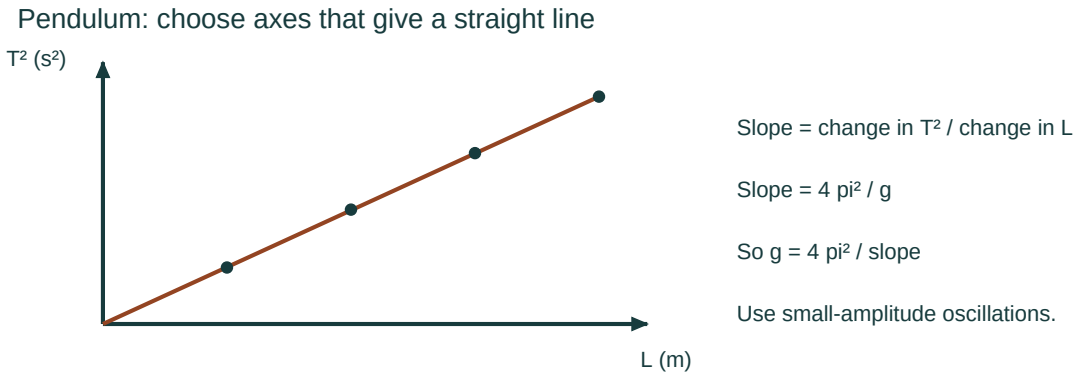
After this lesson, you can

- Measure pendulum length and period consistently.
- Recover g from one trial or a T^2-versus-L graph.

Start with: Complete oscillation; squaring; slope interpretation.

THEORY

For small angles a pendulum period depends mainly on length and gravity, not bob mass. Measure length from pivot to the centre of the bob. Timing several complete oscillations reduces the relative impact of reaction time. Keep the starting angle small and avoid pushing the bob.



Plot period squared vertically against pendulum length horizontally. The theoretical slope is $4\pi^2/g$. Illustrated points are ideal model values, not experimental observations.

RULE

$$T = t/N$$

$$T^2 = (4\pi^2/g)L$$

$$g = 4\pi^2/m \text{ for } T^2 \text{ versus } L$$

L pivot-to-bob-centre length (m)

t time for N complete oscillations (s)

T period of one oscillation (s)

g gravitational acceleration (m/s^2)

When this applies

The simple model assumes small angular displacement, negligible damping and a light inextensible suspension.

A method that earns marks

1. Measure length from pivot to the bob centre and release from a small angle without a push.
2. Time several complete oscillations and divide total time by their count.
3. Calculate T^2 and repeat at several lengths if graphing.
4. For T^2 on the vertical axis and L horizontally, use g = $4\pi^2$/slope.

WORKED EXAMPLE

For L=1 m and T=2 s, g = $4\pi^2$ times 1/4 = 9.87 $\mathrm{m/s^2}$.

1. For the example L = 1 m and T = 2 s.
2. Square the measured period: T^2 = 4 s^2.
3. Rearranging $T^2 = 4\pi^2 L/g$ gives $g = 4\pi^2 L/T^2$.
4. Substitution gives g = $4\pi^2$ times 1/4 ≈ 9.87 m/s^2.

Common mistake
Using total timing t instead of the single-cycle period T makes g dramatically wrong.

Exam habit: State which variables are on which axes before converting a slope into g.

SELF-CHECK

A bob crosses equilibrium twice in one full cycle. Should each crossing be counted as a cycle?

Answer: No. Count a return to the same position moving in the same direction as one full cycle.

LESSON 03 / 12

Free fall and axis choice

After this lesson, you can

- Linearize free fall with a time-squared graph.
- Select the correct g conversion for either axis orientation.

Start with: Constant acceleration from rest; square of time; fitted-line slope.

THEORY

For release from rest with negligible air drag, fall distance grows with time squared. The course sheets mention both graph orientations; the equation must decide the slope conversion. A light sensor or electronic timer avoids large manual timing errors.

RULE

$$h = \tfrac{1}{2} g t^2;$$

$$m_{h \text{ vs } t^2} = g/2;$$

$$m_{t^2 \text{ vs } h} = 2/g$$

h vertical fall distance (m)

t fall time (s)

g gravitational acceleration (m/s^2)

m graph slope (depends on axes)

When this applies

The object starts from rest and air resistance is negligible over the measured fall.

A method that earns marks

1. Release from rest and measure fall time with the stated apparatus.
2. Calculate t^2 for each height.
3. State the graph axes and compare h = (g/2)t^2 with y = mx.
4. For h against t^2 use g = 2m; for t^2 against h use g = 2/m.

WORKED EXAMPLE

A plot of h against t^2 has slope 4.9 m per square second, so g=9.8 $\mathrm{m/s^2}$. Reversing axes gives slope about 0.204 square seconds per metre.

1. With h on the vertical axis, the fitted slope is 4.9 m/s^2.
2. The equation h = (g/2)t^2 shows this slope is half g, so g = 9.8 m/s^2.
3. Reversing the axes makes slope 2/g, about 0.204 s^2/m.
4. The reciprocal slopes describe the same ideal relationship; neither slope can be interpreted without its axes.

Common mistake
Do not use g=2 times slope for a t-squared-versus-height graph.

Exam habit: Do not memorize slope = g/2 without the graph orientation; use the displayed equation to decide.

SELF-CHECK

A t^2-versus-h slope is 0.20 s^2/m. What is g?

Answer: 10 m/s^2, using g = 2/0.20.

LESSON 04 / 12

Static Hooke law

After this lesson, you can

- Distinguish spring extension from loaded length.
- Find spring stiffness from a load or a force-extension slope.

Start with: Weight mg; length differences; centimetres to metres.

THEORY

Within the proportional range, spring extension is proportional to the applied load. Extension is loaded length minus unloaded length. The restoring spring force points opposite the extension, while the applied load magnitude is k times extension. An elastic spring returns to its original length after unloading, but that alone does not prove a straight-line force-extension relation.

RULE

$$F = k\Delta L$$

$$F = mg$$

$$k = m_{F \text{ vs } \Delta L}$$

F applied load magnitude (N)

ΔL loaded minus unloaded spring length (m)

k spring stiffness (N/m)

m total suspended mass (kg)

When this applies

Use Hooke law only over the proportional linear range; elastic recovery alone does not guarantee strict proportionality.

A method that earns marks

1. Record the unloaded spring length first.
2. Add the load safely and calculate extension as loaded minus unloaded length.
3. Convert extension to metres and load mass to force using mg.
4. Find k from F/ΔL or the slope of F against ΔL; check the graph remains linear.

WORKED EXAMPLE

A 2 N load produces 0.04 m extension, so k=50 N/m. A permanent change of unloaded length signals the elastic model may no longer apply.

1. The applied load is 2 N and the extension is 0.04 m, not the total spring length.
2. Using $F = k\Delta L$ gives $k = 2/0.04 = 50$ N/m.
3. This means an additional 1 m extension would require 50 N only if the same linear model remained valid.
4. If unloading leaves a changed length, the spring has not returned elastically; avoid extrapolating the original model.

Common mistake
Centimetres must be converted to metres before reporting k in N/m.

Exam habit: Include the hanger in total suspended mass if the apparatus instructions require it; otherwise the force scale can acquire an offset.

SELF-CHECK

A 0.30 kg total load extends the spring 6 cm with $g = 10$ m/s^2. What is k?

Answer: 50 N/m: the force is 3 N and extension is 0.06 m.

LESSON 05 / 12

Dynamic spring and gravity

After this lesson, you can

- Relate mass-spring period to load and spring stiffness.
- Use graph axes to distinguish a k experiment from a g experiment.

Start with: Static spring extension; period measurement; slope units.

THEORY

An oscillating mass-spring system has a period that increases with mass and decreases with stiffness. Combining static extension with dynamic period allows gravity to be estimated. Small oscillations and the same effective load model are needed for consistency.

RULE

$$T^2 = 4\pi^2 m/k$$
$$T^2 = (4\pi^2/g)\Delta L$$

T oscillation period (s)

m oscillating mass in the stated model (kg)

k spring stiffness (N/m)

ΔL static equilibrium extension (m)

g gravitational acceleration (m/s^2)

When this applies

Small oscillations stay in the spring proportional range; the static and dynamic models must treat hanger and effective spring mass consistently.

A method that earns marks

1. Measure small vertical oscillations about equilibrium and obtain period from several cycles.
2. For T^2 against mass compare the slope with $4\pi^2$/k.
3. For T^2 against static extension compare the slope with $4\pi^2$/g.
4. Apply the matching rearrangement and retain the effective-mass assumptions used by the apparatus.

WORKED EXAMPLE

A T-squared-versus-extension slope of 4.0 square seconds per metre gives g=$4\pi^2$/4.0=9.87 $\mathrm{m/s^2}$.

1. The example plots T^2 vertically against static extension horizontally.
2. Its slope is 4.0 s^2/m, corresponding to $4\pi^2$/g.
3. Rearrangement gives g = $4\pi^2$/4.0 $\approx$ 9.87 m/s^2.
4. A T^2-versus-mass slope would instead determine k; substituting it here would mix two different experiments.

Common mistake
The symbols for mass and graph slope can both be m; label them explicitly to avoid confusion.

Exam habit: Use slope units to identify the requested constant: s^2/kg for stiffness, s^2/m for gravity in these two plots.

SELF-CHECK

A T^2-versus-mass slope is 0.80 s^2/kg. What is k?

Answer: About 49.35 N/m, from $4\pi^2$/0.80.

LESSON 06 / 12

Force table and equilibrium

After this lesson, you can

- Calculate the equilibrant from perpendicular forces.
- Explain how the force-table ring indicates approximate equilibrium.

Start with: Vector components; Pythagoras; angles from positive x.

THEORY

The force table demonstrates vector addition. The resultant of two forces is their vector sum. The equilibrant has equal magnitude and opposite direction. A centred ring that does not press on the pin indicates approximate balance, subject to pulley friction.

RULE

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$F_i = m_i g$$

F_i tension force from hanging mass in the ideal setup (N)

m_i total hanging mass for a string (kg)

R resultant force magnitude (N)

When this applies

Mass-to-tension conversion neglects pulley friction and string mass; actual balance is approximate.

A method that earns marks

1. Convert each hanging mass to a force using the chosen g.
2. Resolve all forces into signed horizontal and vertical components.
3. Find the resultant, then reverse both components for the equilibrant.

4. Adjust the balancing force and check the ring is centred without pressing on the pin.

WORKED EXAMPLE

For 3 N east and 4 N north, the resultant is 5 N northeast. The equilibrant is 5 N southwest.

1. The 3 N east and 4 N north forces give resultant components (+3,+4) N.
2. Its magnitude is $\sqrt{(3^2+4^2)}$ = 5 N.
3. Its precise direction is 53.1° north of east; northeast here describes its quadrant.
4. The equilibrant has components (-3,-4) N and magnitude 5 N, directed 53.1° south of west.

Common mistake
The equilibrant is not the resultant in the same direction.

Exam habit: If an angle is requested, northeast alone is insufficient; give a numerical angle and reference axis.

SELF-CHECK

Does the equilibrant have the same direction as the resultant?

Answer: No. It has equal magnitude and opposite direction.

LESSON 07 / 12

Newton cooling law

After this lesson, you can

- Calculate temperature excess above the surroundings.
- Interpret an exponential cooling trend without confusing it with Celsius temperature.

Start with: Subtracting temperatures; halving; logarithm recognition.

THEORY

A hot object approaches room temperature as its temperature excess decays. The simple model assumes approximately constant surroundings and an effective cooling constant over the measured range. The temperature excess, not the absolute temperature, is the decaying quantity.

RULE

$$T - T_r = (T_0 - T_r)e^{-kt}$$
$$\ln[(T - T_r)/(T_0 - T_r)] = -kt$$

T object temperature (°C or K consistently)

T_r room temperature (same unit as T)

k effective cooling constant (inverse time)

t elapsed time (s or min consistently with k)

When this applies

Room temperature is approximately constant and the effective cooling constant applies over the measured range.

A method that earns marks

1. Measure room temperature and record it as the reference.
2. Subtract room temperature from each object temperature.
3. Compare excesses over time or use the stated logarithmic plot.
4. Add the room temperature back when converting a final excess into an object temperature.

WORKED EXAMPLE

If room temperature is 20 degrees C and the object cools from 80 to 50, the excess falls from 60 to 30 degrees, so it halves.

1. The room is at 20°C, so the initial 80°C object has excess 80 - 20 = 60°C.
2. At 50°C its excess is 50 - 20 = 30°C.
3. The excess has halved, even though 50 is not half of 80.
4. This distinction explains why the model approaches room temperature rather than cooling toward 0°C.

Common mistake
The object approaches room temperature, not zero degrees C.

Exam habit: When logging data, use a dimensionless excess ratio such as (T-Tr)/(T0-Tr), with temperatures consistently measured.

SELF-CHECK

At room temperature 25°C, an object starts at 85°C and its excess halves. What is its new temperature?

Answer: 55°C: initial excess 60°C becomes 30°C, then add 25°C.

LESSON 08 / 12

Boyle law

After this lesson, you can

- Use absolute pressure in a fixed-temperature gas comparison.
- Predict the inverse pressure-volume trend and check thermal settling.

Start with: Inverse proportion; multiplying and dividing; consistent pressure and volume units.

THEORY

At fixed temperature and fixed amount of gas, absolute pressure is inversely proportional to volume. A narrow uniform tube may use gas-column length as a proxy for volume. Let the gas settle thermally after changing pressure.

RULE

$$P_1V_1 = P_2V_2$$

$$P \propto 1/V$$

P absolute gas pressure (Pa or consistent pressure unit)

V gas volume (m^3 or consistent volume unit)

When this applies

Gas amount and temperature remain constant; the ideal Boyle-law approximation is suitable for the measured range.

A method that earns marks

1. Confirm the gas amount stays fixed and readings are taken at the same temperature.
2. Use absolute pressure rather than gauge pressure.
3. Write P1V1 = P2V2 and isolate the requested volume or pressure.
4. Check that larger pressure gives smaller volume and allow time for thermal settling between trials.

WORKED EXAMPLE

A gas at 100 kPa and 60 cubic centimetres has volume 30 cubic centimetres at 200 kPa, if temperature stays fixed.

1. Initially the gas has pressure 100 kPa and volume 60 cm^3.
2. At fixed temperature and amount, their product stays constant.
3. Doubling pressure to 200 kPa requires half the volume: V2 = 100 times 60/200 = 30 cm^3.
4. Keeping kPa and cm^3 is valid for this ratio because the corresponding units match on both sides.

Common mistake
Compressing quickly can warm the gas and violate the constant-temperature assumption.

Exam habit: Do not convert a gauge reading to absolute pressure by ignoring atmospheric pressure; use the apparatus definition.

SELF-CHECK

If volume is compressed quickly, why wait before recording a Boyle-law reading?

Answer: Compression can change temperature; waiting helps the gas return to the intended common temperature.

LESSON 09 / 12

Linear thermal expansion

After this lesson, you can

- Calculate a linear expansion coefficient with compatible length units.
- Extract the coefficient from a length-change-versus-temperature graph.

Start with: Small length changes; temperature differences; mm-to-m conversion.

THEORY

Heating changes the length of a rod slightly. The coefficient of expansion describes fractional length change per degree. A dial gauge often reads millimetres while the rod length is recorded in metres, so consistent units are critical.

RULE

$$\Delta L = \alpha L_0 \Delta T$$

$$\alpha = m_{\Delta L \text{ vs } \Delta T} / L_0$$

α linear expansion coefficient (K^{-1})

L_0 initial rod length (m)

ΔL length increase (m)

ΔT temperature increase (K)

When this applies

The temperature range is small enough to treat α as approximately constant and the rod temperature as sufficiently uniform.

A method that earns marks

1. Record original rod length and the initial and final temperatures.
2. Convert gauge displacement to the same length unit as the rod.
3. Rearrange $\Delta L = \alpha L_0 \Delta T$ to solve for α.
4. For a graph of ΔL against ΔT divide its slope by L_0.

WORKED EXAMPLE

A 1 m rod expands 0.5 mm over 50 K, giving alpha = 0.0005/(1 times 50)=0.00001 per K.

1. The 1 m rod expands by 0.5 mm, which is 0.0005 m.
2. The temperature rise is 50 K.
3. Divide extension by original length and temperature rise: α = 0.0005/(1 times 50).
4. The result is 0.00001 K^{-1}, or 1.0×10^{-5} K^{-1}.

Common mistake
A graph against absolute temperature may have an intercept; do not confuse it with a change-of-temperature plot.

Exam habit: Write the displacement conversion explicitly; the small numerical size of α is physically expected.

SELF-CHECK

What conversion error occurs if 0.5 mm is entered as 0.5 m?

Answer: The calculated expansion coefficient becomes 1000 times too large.

LESSON 10 / 12

Specific heat, latent heat and Joule equivalent

After this lesson, you can

- Separate sensible heating, phase-change energy and electrical input.
- Keep the joule-per-calorie conversion dimensional.

Start with: Energy balance; electrical power; grams-to-kilograms conversion.

THEORY

Mixing calorimetry balances heat lost by the hot sample against heat gained by water and calorimeter. Latent heat accounts for a phase change. Electrical heating provides IVt joules; comparing it with heat measured in calories gives joules per calorie, not a dimensionless ratio.

RULE

$$m_s c_s (T_s - T_f) = (m_w c_w + C_{\text{cal}})(T_f - T_w)$$

$$Q = mL$$

$$E = IVt$$

Q thermal energy transferred (J)

L specific latent heat (J/kg)

I heater current (A)

V heater voltage (V)

t heating duration (s)

When this applies

The latent-heat example assumes the sample starts at its melting point and all stated energy is used for melting.

A method that earns marks

1. Identify whether temperature changes, phase changes or both occur.

2. Calculate electrical input as IVt when heater readings are supplied.
3. Use mcΔT for temperature change and mL for a phase change.
4. If comparing joules with calories, divide joules by calories and state J/cal.

WORKED EXAMPLE

If 500 J melts 0.002 kg at its melting point with no other heating, latent heat is 250000 J/kg. If 418.4 J corresponds to 100 cal, the conversion is 4.184 J/cal.

1. The sample receives 500 J only for melting and the melted mass is 0.002 kg.
2. Rearranging Q = mL gives L = 500/0.002 = 250000 J/kg.
3. In the separate conversion example, 418.4 J is compared with 100 cal.
4. Their ratio is 418.4/100 = 4.184 J/cal; the unit is essential because the two energy scales differ.

Common mistake
Ignoring calorimeter heat capacity or heat loss can bias the inferred material property.

Exam habit: State where the energy goes before dividing by mass; not all heating energy necessarily produces the phase change.

SELF-CHECK

If both electrical input and thermal gain are already in joules, is their ratio a J/cal conversion?

Answer: No. It is dimensionless and tests energy accounting or an efficiency interpretation under the model.

LESSON 11 / 12

Calorimetry: determine a solid specific heat

After this lesson, you can

- Balance heat lost by a solid against heat gained by water and calorimeter.
- Solve for specific heat while using the correct temperature changes.

Start with: Heat capacity; mass in kg; a common final equilibrium temperature.

THEORY

A hot solid placed in cooler water transfers energy to the water and calorimeter. At equilibrium they share a final temperature. The ideal model assumes negligible heat loss to surroundings and no phase change. The calorimeter contribution must be included if it is specified.

RULE

$$c_s = \frac{(m_w c_w + C_{\text{cal}})(T_f - T_w)}{m_s(T_s - T_f)}$$

c_s solid specific heat capacity (J/(kg K))

m_s, m_w solid and water masses (kg)

C_{cal} calorimeter heat capacity (J/K)

T_f final equilibrium temperature (°C or K)

When this applies

The example neglects surroundings losses and calorimeter heat capacity and assumes no phase change.

A method that earns marks

1. Identify the initially hot body, initially cold water and common final temperature.
2. Write heat lost by the solid equal to heat gained by water plus calorimeter.
3. Use positive temperature drops and rises on opposite sides.
4. Solve for cs and state the heat-loss and calorimeter assumptions.

WORKED EXAMPLE

With a negligible calorimeter, 0.10 kg solid cooling from 80 to 30 degrees C heats 0.20 kg water from 25 to 30 degrees C. Taking water c=4200 gives solid c=840 J/(kg K).

1. Water warms from 25°C to 30°C, a rise of 5 K.
2. Its heat gain is 0.20 times 4200 times 5 = 4200 J.
3. The 0.10 kg solid cools from 80°C to 30°C, a drop of 50 K.
4. With negligible calorimeter heat capacity and losses, 0.10 times cs times 50 = 4200, giving cs = 840 J/(kg K).

Common mistake
Using the same temperature change for hot solid and water is generally wrong.

Exam habit: Use the calorimeter heat-capacity term when supplied; discarding it undercounts heat gained by the cold side.

SELF-CHECK

Why is the solid temperature change not the same as the water temperature change?

Answer: They have different starting temperatures, though they reach the same final equilibrium temperature.

LESSON 12 / 12

Latent heat and electrical-to-thermal conversion

After this lesson, you can

- List all energy stages for ice melting and subsequent warming.
- Interpret how lost heat biases a measured electrical-to-calorie ratio.

Start with: Specific heat versus latent heat; phase-change temperatures; units of energy.

THEORY

A phase-change experiment separates melting from subsequent warming of the melted material. In the Joule-equivalent experiment electrical energy is compared with heat gained by water and calorimeter. If both quantities are already in joules, their ratio tests energy accounting; it is not a joule-per-calorie conversion.

RULE

$$Q_{\text{ice}} = mL_f + mc_w(T_f - 0^\circ\text{C})$$

$$J_{\text{eq}} = E_{\text{joules}}/Q_{\text{calories}}$$

m ice mass (kg)

L_f specific latent heat of fusion (J/kg)

c_w liquid water specific heat (J/(kg K))

J_{eq} joules per calorie conversion estimate (J/cal)

When this applies

The ice formulas use an ordinary-pressure melting point of approximately 0°C and account for the stated final phase.

A method that earns marks

1. Identify whether ice starts below or at its melting point.
2. Include warming of ice if needed, then melting, then warming of meltwater to the final temperature.

3. For conversion, keep electrical energy in joules and measured thermal gain in calories.
4. Assess a loss by asking whether the recorded heat gain is too small or too large for the supplied energy.

WORKED EXAMPLE

If ice begins below zero degrees C, an additional warming-of-ice term is required before melting. If it begins at zero, that term is absent.

1. Ice below 0°C first needs energy to warm to 0°C; that stage uses ice specific heat.
2. At 0°C, energy mLf melts it without a temperature rise in the ideal fixed-pressure model.
3. If the final state is liquid above 0°C, add mcw(Tf - 0°C) for the meltwater.
4. When ice already starts at 0°C the first stage is absent; the melting and later warming stages remain if required.

Common mistake
Do not divide joules by joules and label the dimensionless answer J/cal.

Exam habit: Do not use the water specific heat for warming solid ice; the two phases have different heat capacities.

SELF-CHECK

If 836.8 J supplies a recorded heat gain of 200 cal, what is E/Q, and how would unrecorded heat loss bias it?

Answer: 4.184 J/cal. If heat escapes and measured Q is too small, E/Q tends to be too high.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Mechanics and Heat Lab / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. If y is force in N and x is extension in m, slope units are:

   A. N/m

   B. m/N

   C. N times m

   D. No units

2. Pendulum length should be measured to:

   A. The top of the table

   B. The lowest swing point

   C. The bottom of the bob

   D. The centre of the bob

3. For t^2 vertically versus h horizontally, g equals:

   A. Slope squared

   B. Slope/2

   C. 2 times slope

   D. 2/slope

4. The spring constant is the slope of:

   A. Time versus temperature

   B. Extension versus force

   C. Force versus extension

   D. Mass versus time

5. For fixed k, multiplying the oscillating mass by four makes the period:

   A. Four times

   B. Half

   C. Twice

   D. Unchanged

6. The equilibrant of a 7 N eastward resultant is:

   A. 7 N east

B. 14 N west

C. Zero

D. 7 N west

Part II: Show your reasoning

1. A measured g is 9.6 $\mathrm{m/s^2}$ and reference g is 9.8 $\mathrm{m/s^2}$. Calculate percentage error and name one possible systematic error.

2. A pendulum takes 30 s for 15 cycles at length 0.99 m. Calculate period and g.

3. The slope of t^2 against height is 0.20 square seconds per metre. Find g.

4. A 0.30 kg mass extends a spring by 6 cm. Use g=10 $\mathrm{m/s^2}$. Find k.

5. A T-squared-versus-mass graph has slope 0.80 square seconds per kg. Find spring constant.

6. Two forces are 6 N east and 8 N north. Find equilibrant magnitude and direction.

Mechanics and Heat Lab / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. In the simple cooling model, the cooling rate is proportional to:

   A. Mass only

   B. Zero

   C. Temperature excess above surroundings

   D. Absolute time only

2. Boyle law requires:

   A. Fixed temperature and gas amount

   B. Changing gas amount

   C. Fixed pressure

   D. Fixed volume

3. Two rods are made of the same material and have the same temperature rise. A rod twice as long has expansion:

   A. Twice as large

   B. Unchanged

   C. Four times as large

   D. Half as large

4. Electrical energy supplied by constant voltage and current for time t is:

   A. V/I

   B. IVt

   C. I/t

   D. Vt/I

5. In an ideal mixing experiment, the final equilibrium temperature lies:

   A. Below both

   B. Between the initial temperatures

   C. Always at zero

   D. Above both initial temperatures

6. To determine joules per calorie, divide electrical energy in joules by heat measured in:

A. Watts

B. Calories

C. Kelvin

D. Joules

Part II: Show your reasoning

1. Room temperature is 25 degrees C. A body starts at 85 degrees C and its temperature excess halves in 5 minutes. What is its temperature then?

2. A trapped gas has volume 40 cubic centimetres at absolute pressure 120 kPa. Find its volume at 160 kPa at the same temperature.

3. A 0.50 m rod expands 0.30 mm after warming by 40 K. Find its expansion coefficient.

4. A heater at 6 V carries 2 A for 100 s. If all energy melts a sample already at its melting point and the melted mass is 0.004 kg, find supplied energy and latent heat.

5. A 0.20 kg solid cools from 90 to 40 degrees C and warms 0.10 kg water from 30 to 40 degrees C. Ignore calorimeter heat capacity and losses; use water c=4200 J/(kg K). Find the solid specific heat.

6. An electrical experiment supplies 836.8 J and records heat gain of 200 cal. Find the conversion factor. If unaccounted heat escapes, does E/Q measured tend to be too high or too low?

Mechanics and Heat Lab / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **A.** Slope units are vertical units divided by horizontal units.
2. **D.** The ideal mass is represented at the bob centre.
3. **D.** Rearranging h = gt^2/2 gives t^2 = (2/g)h.
4. **C.** Force = k times extension matches y = mx.
5. **C.** Period is proportional to the square root of mass.
6. **D.** Equal magnitude and opposite direction produce zero net force.

Part II: Written solutions

1. Percentage error = 100 times 0.2/9.8 = 2.04 percent. A consistent height-scale zero offset is one possible systematic error. Two marks for the calculation and two for a specific error with its consistency explained.
2. T=30/15=2 s. g=$4\pi^2$ times 0.99/4=9.77 $\mathrm{m/s^2}$. Two marks per calculation.
3. g = 2/0.20 = 10 $\mathrm{m/s^2}$. Four marks for axis identification, rearrangement, substitution and unit.
4. Force = 3 N; extension = 0.06 m; k=3/0.06=50 N/m. Award marks for force, conversion, ratio and unit.
5. Slope = $4\pi^2$/k, so k=$4\pi^2$/0.80=49.35 N/m. Credit the selected axes, rearrangement, numerical value and units.
6. Resultant magnitude is 10 N at 53.1 degrees north of east. Equilibrant is 10 N at 233.1 degrees from positive x, or 53.1 degrees south of west. Two marks for magnitude and two for direction.

Mechanics and Heat Lab / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** Newton cooling uses the difference between object and surrounding temperatures.
2. **A.** Only pressure and volume are varied in the ideal Boyle experiment.
3. **A.** Expansion is proportional to original length.
4. **B.** Electrical power is VI; multiplying by time gives energy.
5. **B.** Energy flows from the hotter body to the cooler one until equilibrium.
6. **B.** The denominator determines the per-calorie unit.

Part II: Written solutions

1. Initial excess = 60 degrees C. Halved excess = 30 degrees C. Add room temperature: 25+30=55 degrees C. Credit both differences and final interpretation.
2. V2 = 120 times 40/160 = 30 cubic centimetres. Credit controlled conditions, equation, rearrangement and unit.
3. Convert expansion to 0.00030 m. Alpha=0.00030/(0.50 times 40)=0.000015 per K. Credit conversion, formula, substitution and result.
4. Energy=6 times 2 times 100=1200 J. Latent heat=120$\frac{0}{0}$.004=300000 J/kg. Two marks per calculation; the no-loss and initial-phase-temperature assumptions matter.
5. Water gains 0.10 times 4200 times 10=4200 J. Solid loses 0.20 times cs times 50=10cs. Thus cs=420 J/(kg K). Two marks for each side and result.
6. Factor=836.8/200=4.184 J/cal. Heat escaping reduces measured Q for the supplied E, tending to make the ratio too high. Two marks for each part.

Source and coverage notes

All distinct experiments in the folder: pendulum, free fall, static and dynamic springs, force table, cooling, Boyle law, expansion, specific heat, latent heat and Joule equivalent. Duplicate versions are consolidated.

Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.